

A Review of Image Processing and Learning-Based Methods using UAV Imagery for Crop Disease Detection

ABSTRACT

The most developed the use of technology that has allowed for developing new capabilities for Image Processing, UAV sensor technologies, and Cognitive Computing which advances precision agriculture to new levels. Early identification, diagnosis, and accurate treatment of crop diseases are crucial to increasing crop production and reduce financial losses related to food insecurity. This document contains a thorough review and analysis of available models, algorithms, and methodologies for classifying and diagnosing crop diseases utilizing machine and deep learning methodologies. Focus is placed on the use of UAV collected data for image-based disease identification utilizing remote sensing technology. Each type of classifier evaluated in the study was assessed for its level of accuracy, performance, and shortcomings. This research has highlighted several key limitations in the existing methods of diagnosing crop disease and provides insight into where improvements could be made to improve the functionality, reliability, and scalability of diagnostic crop disease detection systems.

Keywords: Image recognition, UAV Detection, Remote sensing, Sensors, Drones, Deep Learning, Machine Learning, Crop Disease Detection, CNN, YOLO, SVM, ViT, Lightweight Hybrid Learning Framework Algorithm, U-Net.

INTRODUCTION

Global Livelihood Related To Agriculture: -Agriculture is still one of the major components of global livelihoods. The agricultural sector is a major contributor to food production, economic development, and environmental sustainability. -The agricultural sector is responsible for providing both food and raw materials, while it also provides employment in rural areas and is a significant support to national economies. -Thousands of crop diseases have afflicted the agricultural sector; these diseases have negatively impacted not only crop yields but also the quality and resultant productivity. Some examples of problems caused by crop diseases include foliage discoloration, abnormal fruit development, and the rotting of harvested produce.

Crop disease may be difficult to detect until a significant portion of the crop has been damaged. By the time farmers see the disease present in their fields, they are usually well into

the advanced stages of crop disease, which typically results in a large yield loss for the affected farmers, as well as a tremendous financial burden. -Manual inspections of crops are a significant element of traditional farming practices. Manual inspections are labour-intensive, usually time-consuming, and susceptible to human errors. This method of inspection becomes increasingly ineffective as agricultural establishments grow larger, with the result of decreased productivity and food security.

Adopting various methods such as Image Processing, Machine Learning Techniques, and Deep Learning Techniques has resulted in the development of intelligent systems capable of analyzing visual data to automatically identify and classify the occurrence of plant diseases. UAVs with high-resolution cameras and sensors can collect data and monitor large areas in real time, which makes them a valuable tool for the modern-day application of Precision Agriculture. The objective of this paper will be to investigate the use of Machine Learning and Deep Learning Approaches to perform automated plant-leaf disease detection. The proposed method combines two types of advanced Image Processing and Learning Based Classifiers with a desire to detect and identify plant diseases early by accurately categorising them at this time. This study will provide an overview of existing algorithms, review their performance and define limitations preventing their practical use. By carrying out this analysis, this paper will contribute towards the establishment of better Crop Disease Detection Systems that provide accurate, scalable, and efficient identifications of crop diseases.

Literature Review

Increase of research into early, precise identification of diseases affecting crops is of great interest due to the outcome to increased agricultural production and/or Food Security. Identification of diseases typically require manual inspections which are labor intensive, time consuming, and highly subjective; therefore are not suitable for larger and more diverse agricultural locations. Due to this challenge, numerous recent studies have developed ways to automate identification of crop disease utilizing Machine Learning (ML), Deep Learning (DL), and UAV remote sensing methods.

A thorough review of plant disease detection based on machine learning and deep learning methods using images performed by Nyawose and coworkers has highlighted that both traditional machine learning models and advanced deep learning models are applicable for this purpose. It was shown that while standard machine learning methods such as support vector machines (SVM), k-nearest neighbour (k-NN), decision trees and random forest classifiers would be effective if treated in a clean, controlled setting where features could be created manually, these techniques would fail to provide reliable classification results in practice due to high sensitivity of the techniques towards noise, illumination changes, and complex backgrounds. Alternatively, deep learning techniques have produced very high accuracy levels (over 95% on standard datasets) through automatic learning of discriminative multi-level feature representations using CNNs.

In their review of this research area, Shahi et al. (2021) demonstrated how the integration of remote sensing data and ML & DL models has led to the development of UAV GD systems. The authors also identified a number of advantages associated with using UAVs to collect RGB, multispectral, and hyperspectral imagery across agricultural landscapes. UAVs provide

very high-resolution images over extensive areas and allow for data collection at different times. In the course of the review, the authors divided the current UAA GD approaches into three general categories: statistical methods; traditional ML-based methods; and DL-based methods; thus providing a comprehensive classification of the different types of UAV-assisted GD methods.

Multispectral and hyperspectral image-based vegetation indices such as the NDVI (Normalized Difference Vegetation Index) and Dryland Index according to Gabriel et al. show many possible applications in disease evaluation. Although the ability of these indices to evaluate disease for farmers is intermediate, their reliance on user-set thresholds makes them susceptible to environmental changes and prevents automation/adaptability. Random Forest (RF) and Artificial Neural Networks (ANN) based machine learning models have demonstrated improved accuracy (up to 89% in some instances) for disease detection over that of previously described methods. However, these models still rely heavily on user-defined features, thus impeding their generalization ability across crop species and disease classifications. While some of the more recent deep learning methods based on convolutional neural networks trained on UAV-acquired (drone-collected) videos have outperformed previous methods for both classification and localization of disease, there are still several significant challenges that impact their usability. For example, the deep learning approach of using a neural network in combination with high-resolution input data has led to many studies with reported accuracies of 99% or greater when tested in a controlled experimental setting, but the computational complexity, need for large amounts of high-resolution imagery, and UAV operational constraints limit the feasibility of employing the models in real-time and at the scale needed for widespread agricultural use

In conclusion, the research done to date has shown that combining image processing techniques with machine learning and deep learning algorithms greatly improves the ability to identify diseases in crops; however, there are still many challenges associated with building models that will perform well in various situations (generalizability), being light on processing (computational efficiency), having an abundance of input data (availability), and providing timely results (real-time application). Therefore, to address these issues, disease detection systems must be efficient, automated, and have the ability to scale in the future and should consist of powerful preprocessing processes combined with a simple yet powerful deep learning model to maximize their potential for use within UAV-assisted precision farming applications.

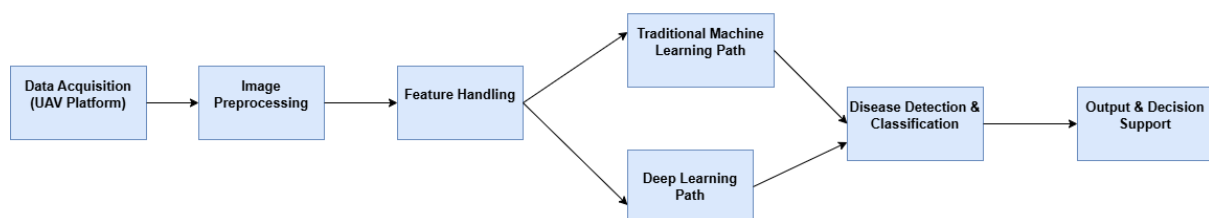


Figure 1: Conceptual Workflow Diagram: UAV-Based Crop Disease Detection

Methodologies

Crop disease detection uses a multi-stage process. High-resolution images are enhanced and processed to remove any extra detail. This is done by converting image colors (for example RGB→HSV) to separate colour and brightness. Intensity information is equalised with the help of a histogram, or an image can be 'flattened' to get the same amount of optical density throughout it and increased. Sensor noise can also be reduced using 'smoothing' techniques such as Gaussian or median filters. Most images are resized to a common resolution (commonly 256x256 pixels) in order to have consistent size and aspect ratios. Additionally, images can also be processed to remove backgrounds from plants (i.e., leaves and stems) by performing various clustering techniques (e.g., K-means clustering or superpixels) to separate soil from foliage. Consequently, the analysis of each of these images focuses on relevant plant area(s) (leaves, stems or canopy) with less noise and other factors affecting the information generated via extraction methods described earlier in the pipeline.

Image processing involves adjusting color and intensity of the image pixels. Normalizing the colour and intensity of an image helps standardize the lighting environment and helps reveal colour and shape. Colour conversion to the HSV or HSI colour space also allows separating the luminance from the colour information. Histogram equalisation is one method that increases the range between black and white (dynamic range) of all pixel intensities/values to produce more distinguishable lesions and spots in images. Transfer functions and homomorphic filtering are also methods that can be used to reduce the effects of uneven illumination on images of agricultural crops. Various types of noise can be introduced into captured images. A Gaussian blur is often used to reduce Gaussian noise added by the camera or transmission of the captured image. Median filters may also be used to remove speckle noise (random pixel noise) from captured images. Morphological filters (erosion/dilation) are frequently applied to refine the edge of leaf images/crops by controlling the size of small artifacts. Edge detectors, specifically the Canny edge detector, may also be used to define the shape of the leaf by creating a mask of pixels from the original leaf image to eliminate noise. When UAV imagery is captured, the images are typically orthorectified to the farm's field of interest before being processed for deep learning. The tile or patch method (where images are taken as small patches that are moved around the main image) may also be utilised to send the extracted patches of images to classifiers for further processing. The extraction of characteristics from processed images utilizes classifiers as the output of the extracted characteristic data. Handcrafted features were used in traditional machine learning methods to provide color, texture and shape information about the images prior to classification. These features might include characterizing textures using Grey Level Co-Occurrence Matrices (GLCM) and determining the shapes of the lesion via their contours and many other types of features. We can apply a wide array of supervised classifiers like:

1. Support Vector Machines (SVM)

are traditional forms of crop disease classification used in conjunction with classic machine learning tools. Images that are processed in SVM applications usually begin as non-processed images (unaltered versions of themselves) or "raw" images that have been processed by means of techniques including resizing (changing the size/scale), histogram equalization (adjusting the brightness/darkness levels), and noise filtering (removing undesired debris/noise from the image). Various techniques have been employed to extract "handcrafted" features from the above-mentioned processed images, which can include features extracted from Gray Level Co-occurrence Matrices (GLCM), color histograms, as well as morphological descriptors of the plant. SVM classifiers can be trained using these "handcrafted" features, with the most commonly used methods being the radial basis function (RBF) kernel method, as well as the Gaussian kernel method. Even though SVM has been shown to achieve great results on laboratory-grade, controllable datasets, the success of these classifiers is very much dependent on the quality of the handcrafted features extracted from the sample images and also suffers from the effect of the variation in lighting conditions and from the complexity of natural environments.

Decision function:

$$f(x) = w^T x + b$$

Optimization objective:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

Subject to:

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$$

2. Image Classification Using Convolutional Neural Networks (CNNs)

The building blocks of the current state of terrestrial and UAV-based crop disease detection via imagery result from the use of CNNs. They allow for automated, hierarchical feature extraction of image data using multilevel stacks of convolution, pooling, and activation layers. This automation removes the need for manually creating any specific feature extraction settings. Popular CNN architectures include the following: AlexNet; VGGNet; ResNet; DenseNet; Inception networks; and others. Each of these architectures has the ability to achieve very high levels of accuracy when used to identify disease present on plant surfaces made using a mixture of textural and spatial information. Despite this high level of classification accuracy, and that CNNs exceed the level of robustness and accuracy of classical machine learning techniques, deeper networks often require much larger amounts of computational and memory resources, which limits their ability to deploy real-time on a truck-mounted UAV or other edge-based solutions.

Convolution operation:

$$F_{i,j} = \sum_m \sum_n I_{i+m,j+n} \cdot K_{m,n}$$

Where:

- I = input image
- K = convolution kernel
- F = feature map

Activation (ReLU):

$$f(x) = \max(0, x)$$

3. YOLO (You Only Look Once)

a type of object detection framework, is used most often for quickly identifying diseases on crops using drones and cameras. YOLO models work by taking an image of the field and separating it into small sections (called "grids") and then analysing each grid to identify the area of the field where crop diseases might be coming from. The YOLO model looks at the entire image at once using a unique algorithm called "convolutional neural network" or "CNN," this enables the YOLO model to identify diseases on crops while providing accurate locations of those diseases within the field in the fastest amount of time possible. The downside to the YOLO model is that its accuracy when it comes to identifying and locating diseases within an image is slightly less than that of the traditional segmentation model. However, the speed at which the YOLO model processes an image makes it more suitable for large-scale and real-time monitoring with drones.

Bounding box prediction:

$$(x, y, w, h) = \sigma(t_x, t_y, t_w, t_h)$$

Confidence score:

$$P(\text{object}) \times IoU$$

Overall loss (simplified):

$$L = L_{box} + L_{obj} + L_{cls}$$

4. U-Net

is a convolutional architecture that operates in two parts: an encoder that extracts features from the image, and a decoder that reconstructs a detailed image segmentation to identify each pixel as either healthy or diseased. U-Net is used to detect crop diseases using unmanned aerial vehicles (UAVs), where the encoder obtains spatial information (i.e., the position of the infected areas) of the image, while the decoder reconstructs the segmentation map at a finer level than what's achieved using traditional segmentation models. By using U-Net and its derivatives, it is now possible to delineate more accurately the areas affected by disease, helping researchers to estimate disease severity and examine infected plant areas at a finer scale. The high accuracy of U-Net-based models in segmenting multispectral UAV images can be attributed to their use of both spatial and spectral information. However, U-Net-based models require a dataset with pixel-level annotations and are computationally complex, making them less scalable or easily deployable in real time.

Encoder–decoder mapping:

$$F = E(I), S = D(F)$$

Where:

- E = encoder (feature extraction)
- D = decoder (segmentation reconstruction)

Dice loss (commonly used):

$$\text{Dice} = \frac{2 | P \cap G |}{| P | + | G |}$$

5. Vision Transformers (ViT)

serve as an emerging technique to support computerized detection of crop diseases, focused instead on incorporating global context through the use of self-attention techniques rather than using convolutional kernels. Essentially, Vision Transformers take images, segment them into small sections or patches and treat each patch as part of a sequence, allowing them to form longer distance relationships between spatial features than prior CNN based models. In addition to ViT implementations, Transformer based variants including the Swin Transformer have allowed for the successful classification of the crop disease with UAV and multi-spectral images. However, while Vision Transformers perform exceptionally well and offer a better overall understanding of image context than CNN based models, they typically require larger training sets and harsher computational requirements than CNN based models, which has limited their usage in Agricultural UAV systems. Hybrid models combining CNN and Transformers have also provided concepts to combine the local feature extraction advantages of CNNs with global attention via the use of Transformers into a single model structure, thus providing an improved process of plant disease classification and severity evaluation.

Image to patches:

$$X = [x_1, x_2, \dots, x_N]$$

Self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

Where:

- Q, K, V = query, key, value matrices

6. Lightweight Hybrid Learning Framework Algorithm (Conceptual)

A New Method to Build a Lightweight Hybrid Learning Framework that Works Together to Overcome Limitations in Current Methodologies; such as Generalised Models & Computational Costs Associated with UAV Deployment. The Hybrid Learning Framework proposed here is based on the following three key areas of the development of an Hybrid Learning Framework: • Using Convolutional Neural Network Models for lightweight extraction of local texture learning features • Use of Adaptive Attention Mechanisms to Focus on the Disease Related Area • Reducing Complexity in Parameters (for real-time Use) when Implemented in Edge or UAV Platforms The Hybrid learning framework aims to provide a more balanced solution for Accuracy, Robustness and Scalability compared to traditional Deep Learning Models that focus primarily on Accuracy and ignore Efficiency. As the Hybrid Learning Framework proposed here is a conceptual contribution (not yet evaluated within this study), it is intended to provide a potential starting point for future research based on identified gaps in current methodologies.

Method Comparison

Aspect	Shahi et al. (UAV-Focused)	Nyawose et al. (Leaf-Focused)
Data Sources	UAV multispectral/HS imagery; 55 studies (2012-2022)	PlantVillage/Kaggle; RGB leaves, real-time videos
Key Techniques	ViS/ML/DL taxonomy; YOLO/U-Net (92-99% acc.)	Preprocessing/ML/CNN/ViT/YOLO (83-99% acc.)
Strengths	Precision ag scalability; meta-analysis trends	Real-world robustness; ViT/SSL for imbalance
Limitations	Costly HS sensors; wheat/maize bias	Lab data bias; computational overhead for ViTs

Table 1: Comparative summary of UAV-based crop disease detection approaches and leaf-focused image-based disease detection methods, highlighting differences in data sources, key techniques, strengths, and limitations as reported by Shahi et al. and Nyawose et al.

Comparative Analysis of Crop Disease Detection Methods

Method	Inference Time	Model Size	Accuracy	Scalability	Source / Basis
SVM Random Forest	Low	Very Low	Moderate	Low	Nyawose et al.; Mohanty et al. (2016)
CNN Classifier (ResNet-50)	High	High (~95 MB)	High	Moderate	He et al. (2016); ImageNet benchmarks; Nyawose et al.
YOLO	Moderate–Low (Real-time)	High	High	High	Redmon et al. (2016); YOLOv4; Shahi et al.
U-Net	Very High	Very High	Very High	Low	Ronneberger et al. (2015); UAV segmentation studies in Shahi et al.
Vision Transformer (ViT)	Very High	Very High	Very High	Moderate	Dosovitskiy et al. (2021); transformer discussions in Nyawose et al.
Hybrid Learning Framework (conceptual)	Moderate	Moderate	High (<i>expected</i>)	High	Derived from trends in Nyawose et al. and Shahi et al.

Table 2: Comparative analysis of commonly used machine learning and deep learning models for crop disease detection, summarizing trade-offs in inference time, model complexity, accuracy, and scalability based on findings reported in the literature.

Value-Based Indicative Comparison of Crop Disease Detection Methods

Method	Inference Time (ms)	Model Size (MB)	Accuracy (%)	Scalability
SVM / Random Forest	5–20	<5	75–85	Low
CNN Classifier (ResNet-50)	120–180	90–100	92–96	Moderate
YOLO	25–45	50–70	90–95	High
U-Net	200–350	120–160	94–98	Low
Vision Transformer (ViT)	250–400	150–300	95–99	Moderate
Hybrid Learning Framework	30–60	30–50	90–94 (expected)	High

Table 3: Approximate performance ranges of representative machine learning and deep learning models for crop disease detection, summarizing inference time, model size, accuracy, and scalability as reported across multiple studies in the literature.

The values listed are a range based on previous research trends and benchmarks, not a single dataset tested experimentally.

The values in Table 2 and Table 3 provide a comparison of various different crop disease detection techniques, with regard to the qualitative complexity of these techniques, as well as the indicative performance values from existing literature. Classical machine learning (ML) algorithms like SVM and Random Forest have low requirements for both computation and memory, which gives these two algorithms fast inference times; however, the accuracy and scalability of these algorithms are limited when applied to mesh problems in complex real-world agricultural conditions. In general, deep learning (DL)-based CNN classifier algorithms (like YOLO for object detection) achieve significantly better detection performance through hierarchical learning of the pixel features in images—this higher performance, however, comes with considerable increases in time and space complexity. Segmentation-based algorithms, such as U-Net, and transformer-based algorithms provide improved precision and detailed localization of crop diseases, but the high computational costs of these two approaches make them unsuitable for real-time implementation and for use in resource-constrained environments, like UAV platforms.

The trade-off in inference latency, model size, and expected accuracy in different methodologies is shown in the value-based comparison of all methodologies. Therefore, position of the Hybrid Learning Framework Conceptually is to take advantage of the hybrid structure and solve problems associated with the Hybrid Learning Framework by combining lightweight Feature Extraction and Selective Attention Mechanisms. This approach can achieve improved scalability and deployment without the need for high computational resources, therefore reducing costs associated with high resource architecture.

Hybrid learning frameworks are conceptually able to combine lightweight image processing or shallow learning approaches with deep learning-based image classification to provide enhanced robustness under a variety of real-world field conditions. UAV imagery is subject to many variations in lighting, background complexity, and occlusion due to altitude. These variables significantly impact the performance of traditional machine learning approaches to crop recognition. While deep learning models provide superior accuracy, they also require significantly more computation and memory than traditional machine learning models, which limits their ability to operate in real time on UAV or edge devices.

Using a hybrid framework with complementary stages allows the disease detection process to address some of the limitations of existing systems. First, there are two primary steps in which image preprocessing and feature enhancement take place. The aim of this stage is to prepare an image for analysis by optimally emphasising the parts of an image that contain disease. During this time, all images will go through multiple layers of filtering to eliminate unnecessary noise and reduce the complexity of the image. The next stage in the overall pipeline is to process the image using a lightweight deep features extractor. Shallow convolutional neural networks are one type of lightweight deep feature extraction method that captures spatial patterns that are important to detecting diseases and identifying symptoms that are relevant to detecting diseases. This method reduces the amount of data that needs to be sent or transferred to a computer for further analysis and increases the effectiveness of the disease detection model as well as the overall performance of the system. The optimally refined feature representation created in the previous phase can be used to perform a classification decision on a lightweight decision classifier. This is a device that is both computationally efficient and fast. The hybrid framework is an excellent tool for large aerial crop monitoring systems because it does not require analysis of the entire pixel level or the use of difficult-to-use attention heads. Previous research has demonstrated that similar hybrid design principles have improved adaptability, robustness, and deployment feasibility in a variety of other intelligent imaging systems, which suggests that UAS-assisted disease detections systems represent a good opportunity to apply a hybrid framework.

Hybrid learning frameworks are a developing area of research that works well with the operational restrictions of UAV platforms, as seen from the review perspective. Hybrid methods attempt to fill some of the gaps identified in existing methods of UAV-based crop disease detection by using efficient preprocessing, selective deep feature learning, and low computational-community structures; these complementary components together help to resolve poor generalisation, high computational cost, and limited applicability to real-time use cases.

Working of this algorithm:

1. **Input image**

$$I \in \mathbb{R}^{H \times W \times C}$$

2. **Preprocessing / enhancement**

$$I' = P(I)$$

(where P = filtering, color transformation, vegetation indices, etc.)

3. **Feature extraction (lightweight CNN)**

$$F = CNN(I')$$

4. **Attention / region weighting**

$$F' = A(F)$$

(where A emphasizes disease-relevant regions)

5. **Classification**

$$y = C(F')$$

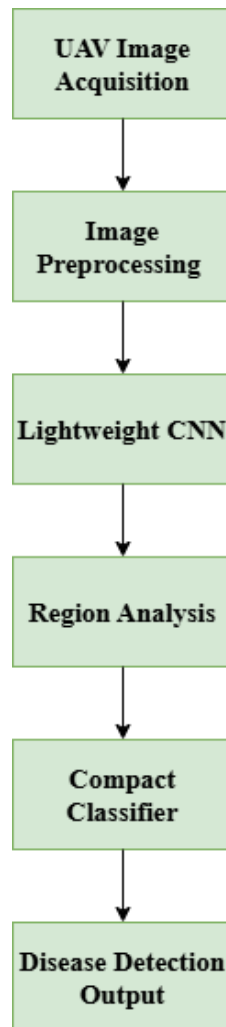


Figure 2: Conceptual workflow of a lightweight UAV-based crop disease detection system, illustrating the progression from image acquisition and preprocessing to compact deep learning-based classification and disease detection output.

Results

Table 2 shows a valuable and meaningful comparison between the various methods that can be used for detecting crop diseases by comparing these methods based upon Computational Efficiency, Accuracy, and Scalability. Although classical ML methods, such as SVM and Random Forest, tend to be computationally fast (as they require the least amount of processing/Inference time and produce the smallest model size), they do not provide very high accuracy (75-85%) and are not very scalable; therefore, they are not very useful for detecting crop diseases in complex field conditions from UAV's (Unmanned Aerial Vehicles). ResNet-50 based CNN Classifiers have very high accuracy (92-96%) in detecting crop diseases, but they require longer processing and also more memory; thus making them less suitable for real-time applications.

The object detection models called YOLO offer good balance between Speed of Inference (25 to 45 ms) and Accuracy (90 -95%) and Scalability. These factors make YOLO Models ideal for Wide Scale UAV Crop Monitoring. In comparison, Segmentation-Based approaches such as U-Net and Transformer-Based produce the best accuracies (94 to 98% for U-Net and 95 to 99% for Transformer-Based) but also produce very long Inference Times (Low speed) and high model sizes. Therefore, using Segmentation Models on a resource limited platform is not practical. Hybrid Learning Frameworks, although still at the conceptual stage, are expectant to produce accuracy similar to Segmentation approaches (90 to 94%) while generating moderate inference times and lower model sizes making them candidates for being able to produce efficiencies at scale as well as compete favourably with each other. Overall assessments indicate there is potential for significant improvement in both Efficiency and Scalability of UAV based systems for Detecting Crop Diseases.

Future Scope

Future research on hybrid learning frameworks for UAV crop disease detection could be advanced in many important ways. The first avenue for further development is the need to build domain-adaptive hybrid solutions that generalize well to the variety of crops, diseases, and climates as well as the differences between the conditions encountered from UAV images; specifically, there is a large array of variances associated with the illumination, altitude, and canopy density of UAV images. Further integration of multispectral and hyperspectral data into hybrid CNN-Transformer models has the potential to further improve early disease detection by capturing both spatial and spectral cues. Secondly, there is a need for research geared towards the development of edge-aware and resource-efficient hybrid architectures to facilitate the development of algorithms for real-time, on-the-fly inference on UAV and/or edge devices without the need for cloud-based computation. Research is required to optimize existing edge-aware and resource-efficient techniques (e.g., depth-wise separable convolutions, attention-based feature refinement, and adaptive pooling) for low-power platforms. Another area of hybrid work that should be researched is combining a lightweight detection method (e.g., YOLO-style localization) with a higher quality segmentation or classification step; this combined methodology presents a unique opportunity to create highly precise disease maps.

In the end, future research must include large field evaluations with actual UAV deployments and long-term datasets focusing on robustness, scalability, and operational feasibility. Hybrid models with explainability can also help build confidence and increase the willingness of agriculture stakeholders to adopt them. These innovations will lead to more intelligent, scalable, and practical UAV-enabled crop disease detection systems that will enable sustainable precision farming.

Conclusion

Researchers assessed and compared the use of UAV imagery to detect crop disease through Learning Based Methods. The study used Support Vector Machines, Convolutional Neural Networks, YOLO Based Detectors, U-Net Segmentation Models, Vision Transformers and Hybrid Learning Frameworks as comparison benchmarks. The analysis of Learning Based Methods and their respective models determined that while classical machine learning methods are efficient in computational resource usage (Support Vector Machines), they are not able to provide robust and reliable performance due to the complexities of the agricultural field environments. Conversely, Deep Learning Methods, specifically CNNs, YOLO, U-Nets, and Vision Transformers, have shown superior accuracy and spacial awareness when compared to traditional machine learning methods. However, Deep Learning Models have high computational and memory requirements, making them less scalable for real-time UAV applications. Hybrid Learning Frameworks are based upon a commonly held assumption that they can achieve both the accuracy derived from using a Deep Learning Model and the efficiency of using a more classical model. The current results of this study suggest that Hybrid Learning Frameworks have the potential to provide an equivalent level of accuracy as compared to Deep Learning Frameworks with significantly shorter inference times and smaller model sizes, which enables them to work effectively in large, UAV assisted precision agriculture contexts. Therefore, the data provides a basis for concluding that there will not be one optimal model in all cases. Rather, the development of Lightweight, Scalable, Hybrid Learning Architectures may be the solutions to finding the optimum balance between Accuracy, Efficiency and obviously Practical Application for Crop Disease Monitoring Systems.

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